Question 1:

Let $f(x) = x^2 + 4$ and $g(x) = x^3 - 3x - 5$

a) $f(x) + g(x) =$ ___?___

b) $D_x[f(x) + g(x)] =$ ___?___

c) $D_x[f(x)] + D_x[g(x)] =$ ___?___

Question 2:

Let $f(x) = x^2 - 2$ and $g(x) = x^3 - x - 1$

a) $3 \cdot f(x) + 8 \cdot g(x) = \underline{\quad ? \quad}$

b) $D_x\left[3 \cdot f(x) + 8 \cdot g(x)\right] = \underline{\quad ? \quad}$

c) $3 \cdot D_x\left[f(x)\right] + 8 \cdot D_x\left[g(x)\right] = \underline{\quad ? \quad}$

d) $D_x\left[a \cdot f(x) + b \cdot g(x)\right] = \underline{\quad ? \quad} \cdot D_x\left[f(x)\right] + \underline{\quad ? \quad} \cdot D_x\left[g(x)\right]$

Question 3:

Let $f(x) = x^4 + 7$

a) $3 \cdot f(x) =$ ___?___

b) $\left[3 \cdot f(x)\right]' =$ ___?___ $\cdot \left[f'(x)\right]$

c) $\left[3 \cdot f(x)\right]'' =$ ___?___ $\left[f''(x)\right]$

d) $\left[3 \cdot f(x)\right]''' =$ ___?___ $\left[f'''(x)\right]$

e) $\left[3 \cdot f(x)\right]^{(n)} =$ ___?___ $\left[f^{(n)}(x)\right]$

f) $\left[a \cdot f(x)\right]^{(n)} =$ ___?___ $\left[f^{(n)}(x)\right]$

Question 4:

Find the general solution of the differential equation $y'' - 3y' - 10y = 0$.

Characteristic or indicial equation: $m^2 - 3y - 10 = 0$.

a) Roots of the characteristic equation are: $m_1 = \underline{\quad ? \quad}$ and $m_2 = \underline{\quad ? \quad}$.

b) Is the function $y_1(x) = e^{m_1 \cdot x}$ a solution of the differential equation $y'' - 3y' - 10y = 0$?

c) Is the function $y_2(x) = e^{m_2 \cdot x}$ a solution of the differential equation $y'' - 3y' - 10y = 0$?

d) Are the functions $y_1(x) = e^{m_1 \cdot x}$ and $y_2(x) = e^{m_2 \cdot x}$ linearly independent?

e) The general solution of the differential equation $y'' - 3y' - 10y = 0$ is $\underline{\quad ? \quad}$.

Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

Question 5:

Second-order differential equation: $2y'' - 13y' + 15y = 0$.

Characteristic or indicial equation: $2m^2 - 13m + 15 = 0$.

a) The roots of the indicial equation $2m^2 - 13m + 15 = 0$ are $m_1 =$ ___?___ and $m_2 =$ ___?___.

b) Is the function $y_1(x) = e^{m_1 \cdot x}$ a solution of the differential equation $2y'' - 13y' + 15y = 0$?

c) Is the function $y_2(x) = e^{m_2 \cdot x}$ a solution of the differential equation $2y'' - 13y' + 15y = 0$?

d) Are the functions $y_1(x) = e^{m_1 \cdot x}$ and $y_2(x) = e^{m_2 \cdot x}$ linearly independent?

e) The general solution of the differential equation $2y'' - 13y' + 15y = 0$ is ___?___.

Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

Question 6:

Second-order differential equation: $3y'' - 19y' - 14y = 0$.

Characteristic or indicial equation: $3m^2 - 19m - 14 = 0$.

a) The roots of the indicial equation $3m^2 - 19m - 14 = 0$ are $m_1 =$ ___?___ and $m_2 =$ ___?___.

b) Is the function $y_1(x) = e^{m_1 \cdot x}$ a solution of the differential equation $3y'' - 19y' - 14y = 0$?

c) Is the function $y_2(x) = e^{m_2 \cdot x}$ a solution of the differential equation $3y'' - 19y' - 14y = 0$?

d) Are the functions $y_1(x) = e^{m_1 \cdot x}$ and $y_2(x) = e^{m_2 \cdot x}$ linearly independent?

e) The general solution of the differential equation $3y'' - 19y' - 14y = 0$ is ___?___.

Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

Question 7:

Second-order differential equation: $9y'' + 12y' + 4y = 0$.

Characteristic or indicial equation: $9m^2 + 12m + 4 = 0$.

a) The roots of the indicial equation $9m^2 + 12m + 4 = 0$ are $m_1 =$ ___?___ and $m_2 =$ ___?___.

b) Is the function $y_1(x) = e^{m_1 \cdot x}$ a solution of the differential equation $9y'' + 12y' + 4y = 0$?

c) Is the function $y_2(x) = x \cdot e^{m_2 \cdot x}$ a solution of the differential equation $9y'' + 12y' + 4y = 0$?

d) Are the functions $y_1(x) = e^{m_1 \cdot x}$ and $y_2(x) = x \cdot e^{m_2 \cdot x}$ linearly independent?

e) The general solution of the differential equation $9y'' + 12y' + 4y = 0$ is ___?___.

Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

Question 8:

Second-order differential equation: $16y'' - 8y' + y = 0$.

Characteristic or indicial equation: $16m^2 - 8m + 1 = 0$.

a) The roots of the indicial equation $16m^2 - 8m + 1 = 0$ are $m_1 = \underline{\quad ? \quad}$ and $m_2 = \underline{\quad ? \quad}$.

b) Is the function $y_1(x) = e^{m_1 \cdot x}$ a solution of the differential equation $16y'' - 8y' + y = 0$?

c) Is the function $y_2(x) = x \cdot e^{m_2 \cdot x}$ a solution of the differential equation $16y'' - 8y' + y = 0$?

d) Are the functions $y_1(x) = e^{m_1 \cdot x}$ and $y_2(x) = x \cdot e^{m_2 \cdot x}$ linearly independent?

e) The general solution of the differential equation $16y'' - 8y' + y = 0$ is $\underline{\quad ? \quad}$.

Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

f) Find the particular solution corresponding to initial conditions of $y(0) = 1$ and $y'(0) = 5$.

Question 9:

Second-order differential equation: $y'' + 5y' + 10.25y = 0$.

Characteristic or indicial equation: $m^2 + 5m + 10.25 = 0$.

a) The roots of the indicial equation $m^2 + 5m + 10.25 = 0$ are $m_1 =$ __?__ and $m_2 =$ __?__. __?__.
   Note: Roots will be complex numbers of the form $m_1 = \alpha + \beta i$ and $m_1 = \alpha - \beta i$

b) Is the function $y_1(x) = e^{\alpha \cdot x} \cdot \cos(\beta x)$ a solution of the differential equation $y'' + 5y' + 10.25y = 0$?

c) Is the function $y_1(x) = e^{\alpha \cdot x} \cdot \sin(\beta x)$ a solution of the differential equation $y'' + 5y' + 10.25y = 0$?

d) Are the functions $y_1(x)$ and $y_2(x)$ linearly independent?

e) The general solution of the differential equation $y'' + 5y' + 10.25y = 0$ is __?__.
   Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

Question 10:

Second-order differential equation: $3y'' + 5y' + 10y = 0$.

Characteristic or indicial equation: $3m^2 + 5m + 10 = 0$.

a) The roots of the indicial equation $3m^2 + 5m + 10 = 0$ are $m_1 =$ ___?___ and $m_2 =$ ___?___.

   Note: Roots will be complex numbers of the form $m_1 = \alpha + \beta i$ and $m_1 = \alpha - \beta i$

b) Is the function $y_1(x) = e^{\alpha \cdot x} \cdot \cos(\beta x)$ a solution of the differential equation $3y'' + 5y' + 10y = 0$?

c) Is the function $y_1(x) = e^{\alpha \cdot x} \cdot \sin(\beta x)$ a solution of the differential equation $3y'' + 5y' + 10y = 0$?

d) Are the functions $y_1(x)$ and $y_2(x)$ linearly independent?

e) The general solution of the differential equation $3y'' + 5y' + 10y = 0$ is ___?___.

   Hint: The general solution has the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$.

f) Find the particular solution corresponding to initial conditions of $y(\pi) = 1$ and $y'(\pi) = 0$.